

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 5 Lecture - Second-order Derivatives and Root Finding

Last Week...

- Finding **maximum**, **minimum** or **saddle point** of a function, or optimising a function, requires a number of steps:
- We see that for a function $f(x)$:
  - If $f'(x) > 0$ then $f(x)$ is increasing.
  - If $f'(x) < 0$ then $f(x)$ is decreasing.
  - If $f'(x) = 0$ it is likely that $f(x)$ will be changing direction!
- Once we know where the these stationary values are for a function, further analysis can help classify them.

Last Week...

- Finding stationary values requires a number of steps:
  1. Differentiate $f(x)$ to find the gradient function $f'(x) = \frac{dy}{dx}$
  2. Set the derivative to zero, $\frac{dy}{dx} = 0$ and solve for x
  3. If necessary, we substitute the stationary values, x, into the function, $f(x)$.
- Once you find where the derivatives are equal to 0 you can explore regions either side to determine if the gradient is positive or negative, and where the function is increasing or decreasing.
- This can tell us whether the stationary point is a maximum, minimum and/or inflection point. But, there is a more formal way to do this!

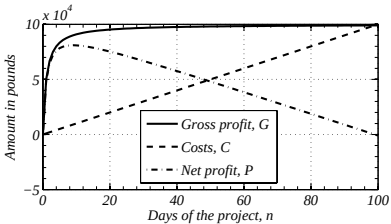
Last Week...

- Finding stationary points (maximum or minimum) proves useful to optimise functions or processes.
- Last week, we saw an example where we wanted to maximise profits from gold mining.
- The net profit from mining, P, as a function of days, n, was given by:

$$P(n) = G(n) - C(n) = 100 - \frac{100}{n+1} - n$$

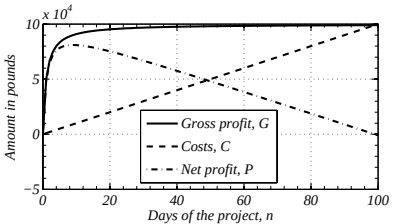
- To find the day where we have maximised (i.e. optimise) our profit, we look for stationary values

$$\frac{dP}{dn} = \frac{100}{(n+1)^2} - 1 = 0$$



Last Week...

- Rearranging this equation showed us that there are stationary values at $n = -11$ and $n = 9$. We ignore $n = -11$ as we cannot have negative time!
- We can see from the graph that at $n = 9$, P is maximised!
- Therefore, we optimise the profit at 9 days so we should stop mining after that.
- We can see that this is a maximum from the plot, but we could/should check this.



Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Explain the concept and significance of second-order derivatives in the analysis of functions.

(2) Compute second-order derivatives of functions to classify stationary values as maxima, minima, or saddle points in optimisation problems.

(3) Describe steady-state systems in physical and mathematical modeling.

(4) Apply Newton's Method to find the roots of complex functions in numerical problem-solving.

Second-order Derivatives

- Differential equations have an "order".
- Up to this point, we have only differentiated functions, $f(x)$ once:

$$f'(x) = \frac{df}{dx}$$

- This is known as the **first-order derivative**.
- Differentiation can be carried out multiple times.
- By differentiating again, we obtain the **second-order derivative** which is defined as:

$$f''(x) = \frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d^2f}{dx^2}$$

- For example, later in the course you will solve differential equations of the form:

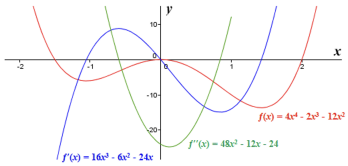
$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

Second-order Derivatives

- The second-order derivative has some very interesting and useful properties.
- Previously we saw that the first-order derivative, $f'(x)$ tells us about the rate of change of a function, $f(x)$.
- The second-order derivative, $f(x)$, goes a step further by telling us how the rate of change, $f'(x)$, itself is changing.
- You can find a cars acceleration by finding out the second derivative (i.e. the rate of change of speed with respect to time) of the distance travelled with respect to the time):
  - Distance travelled at time t: $x = f(t)$.
  - Velocity at time t: $v = f'(t)$.
  - Accelaration at time t: $a = f''(t)$

Second-order Derivatives

- Second-order derivatives have interesting properties and can be used to confirm whether a stationary point is a maximum or minimum point!
- We see that for a stationary value x for the function $f(x)$:
  - If $f''(x) < 0$ the stationary value is a maximum
  - If $f''(x) > 0$ the stationary value is a minimum
  - If $f''(x) = 0$ it is not possible to conclude anything about the stationary value.
- If we're in the situation that $f''(x) = 0$, then it is potentially a saddle point. Need to investigate using the properties of the first-order derivative.



Example: Quadratic function

- Last week, we found the stationary value of the following quadratic:

$$f(x) = \frac{x^2}{2} - 5x + 2$$

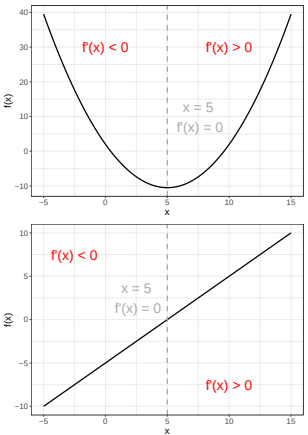
- We did this by differentiating and setting to zero

$$\frac{df}{dx} = x - 5 = 0 \implies x = 5$$

- We found a stationary value at $x = 5$ which we can see is a minimum. Confirm it by looking at the second-order derivative:

$$\frac{d^2f}{dx^2} = 1$$

- Since $\frac{d^2f}{dx^2} > 0$ then we confirm that $x = 5$ is a minimum



Example: Cubic function

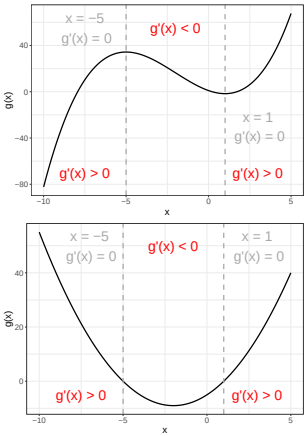
- Last week, we found the stationary value of the following cubic:

$$f(x) = \frac{x^3}{3} + 2x^2 - 5x + 1$$

- We did this by differentiating and setting to zero:

$$\frac{df}{dx} = x^2 + 4x - 5$$
$$\frac{df}{dx} = 0 \implies (x-1)(x+5) = 0$$

- The function has a stationary points at $x = -5$ and $x = 1$ which we see are maximum and minimum respectively.



Example: Cubic function

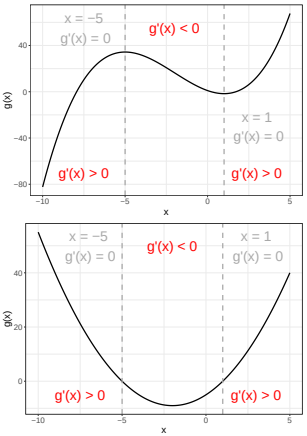
- Confirm if it is by looking at the second order derivative:

$$\frac{df}{dx} = 2x + 4$$

- Evaluate the second order derivatives at the stationary points:

$$\frac{df}{dx}(-5) = 2 \times (-5) + 4 = -6 \qquad \frac{df}{dx}(1) = 2 \times 1 + 4 = 6$$

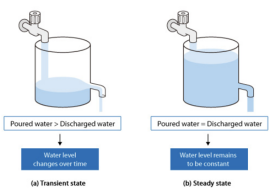
- Since $\frac{d^2f}{dx^2} < 0$ at $x = -5$ then we confirm it is a maximum.
- Since $\frac{d^2f}{dx^2} > 0$ at $x = 1$ then we confirm it is a minimum.



Steady State

- You may not be familiar with the concept of **steady-state**.
- When a system is in steady-state, the key variables that describe the system's behavior do not change over time.
- In a chemical system, we might observe that the systems variables remain constant despite ongoing processes that would lead to change. This requires there is flow through the system.
- Consider the situation where water is poured into a container which has an outlet part on the side of the container.

Comparison	Steady State	Equilibrium
Definition	Used to define consecutive reaction changes	Used to define reversible reaction changes
General equation	$A \xrightarrow{k_1} B \xrightarrow{k_2} C$ $k_1 = -k_2$	$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$ $k_1 = k_2$
Entropy change	Entropy increases with time	Entropy remains constant
Type of system	Open system	Closed system
Heat transfer	Transfer of heat occurs	No net transfer of heat occurs
Type of process	Kinetically controlled processes	Thermodynamically controlled processes

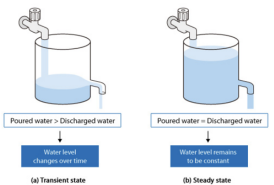


Steady State

- If a system is in steady-state, the variables that define the behaviour of the system do not change with time. What does this mean mathematically?
- We can say that the rate of change of the water volume remains constant, either with respect to time or space.
- Suppose the rate of change can be written as:

$$\frac{dV}{dt} = f(t, x_1, x_2)$$

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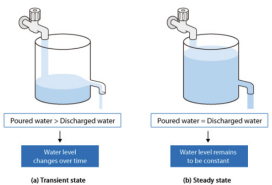


Steady State

- This equation describes the rate at which the total volume of water V is a function of time and multiple sources and sinks of water.
- Using the definition of steady state, the rate of change is constant over time, we can take the derivative again to show that:

$$\frac{d^2V}{dt^2} = 0$$

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Example: Car travelling at constant speed

- Imagine you are in a car that is travelling down the motorway.
- The equation describing the distance travelled (in miles) by time is given by:

$$x(t) = 1 + 68x$$



- Is this system in steady state?
- Let's take the first- and second-order derivative:

$$\frac{dx}{dt} = 68 \qquad \frac{d^2x}{dt^2} = 0$$

- As $\frac{dx}{dt}$ (the velocity) is constant and $\frac{d^2x}{dt^2}$ (the acceleration) is zero, then the car is in steady state!

Root Finding

- We keep discussing the importance of derivatives to understand change and most of our examples have been a combination of hypothetical and real-world examples designed to calculate the gradient of a line.
- One application involves finding the roots of a function: calculating when a function has a value of 0.
- In the examples you have seen, you can factorise this out easily, by rearranging an equation, or by hand.
- In many cases this will not be possible. For example, how do you solve:

$$2x^2 + 5 - e^x = 0$$

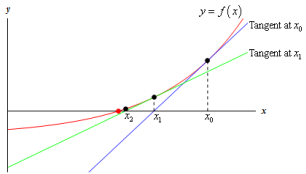
- In this case, it's difficult to find where $f(x) = 0$ using simple algebra.

Root Finding

- We could look at the plot and iteratively look for values until we get close to $f(x) = 0$
- Time consuming and laborious, but also impossible to visualise for complex functions.
- What if we could somehow use the trajectory information provided by gradients, thus tangent lines, in some way?
- Imagine our first value for x is denoted as x_0. We see that $f(x_0) > 0$ and we have a gradient (and tangent line),

$$\frac{dy}{dx}(x_0) = f'(x_0)$$

- We can use this to follow the tangent line from the curve to the x-axis.

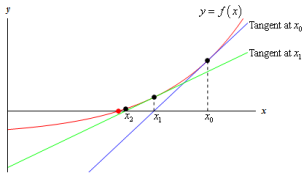


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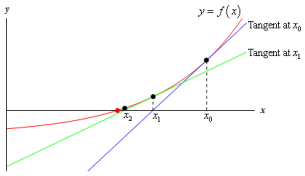
Root Finding

- The tangent crosses the x-axis and arrives at another value for x which we shall denote x_1.
- We can write this as an equation:

$$y = f(x_0) + f'(x_0)(x_1 - x_0)$$

- We can see on our graph that when, in this equation, $y = 0$, then $x = x_1$. So we can find our value for x_1 by setting $y = 0$ and re-arranging:

$$\begin{aligned} y &= f(x_0) + f'(x_0)(x_1 - x_0) \\ x_1 - x_0 &= -\frac{f(x_0)}{f'(x_0)} \\ x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \end{aligned}$$

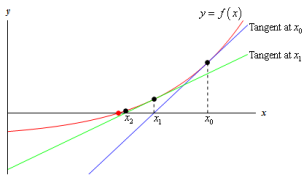


Root Finding

- This leads us to Newtons Method for Root Finding:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- We do this iteratively and we can see on the right this gets us closer and closer to the root.
- You may never reach an exact value for $f(x) = 0$ so we pick a threshold.
- For example, if y is less than 0.001 or values for x_n do not change by more than 0.01%.
- This depends on the nature of the problem you are trying to solve.



Example: Root Finding

- Let's consider the example where we want to find roots of:

$$f(x) = 2x^2 + 5 - e^x$$

- To use Newtons Method, we take the derivative:

$$f'(x) = 4x - e^x$$

- To find a root, we need an initial guess. This can be important. But for now, let's start at $x_0 = 3.5$:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3.5 - \frac{3.6154}{19.1154} = 3.3109$$

- In the weekly exercises, you will be asked to proceed with this problem until you arrive at an approximation for the root which is $x \approx 3.275601$.

